QR Code and Barcode Detection System

Image Processing Module | Spring 2025–2026

Central Asian University | Image Processing | Ismoil Rahimov | 230514

Presentation Outline

What Are QR Codes & Barcodes?

QR Code (Quick Response) — a 2D matrix barcode that stores data in a grid of black and white squares. Created by Denso Wave (Japan, 1994).

Barcode — a 1D optical label consisting of parallel lines encoding numbers and characters (EAN-13, Code 128, etc.).

Both are decoded by cameras and optical scanners, enabling fast data retrieval without manual entry.

💳 Payments

🎫 Ticketing

📦 Logistics

🔗 Web links

Problem Statement & Objectives

Given a digital image or live video — the system must:

🔍

Locate

Find all QR / barcodes in the scene

Locate

📖

Decode

Convert binary pattern → readable text

Decode

🖼️

Visualise

Draw bounding boxes + overlay decoded text

Visualise

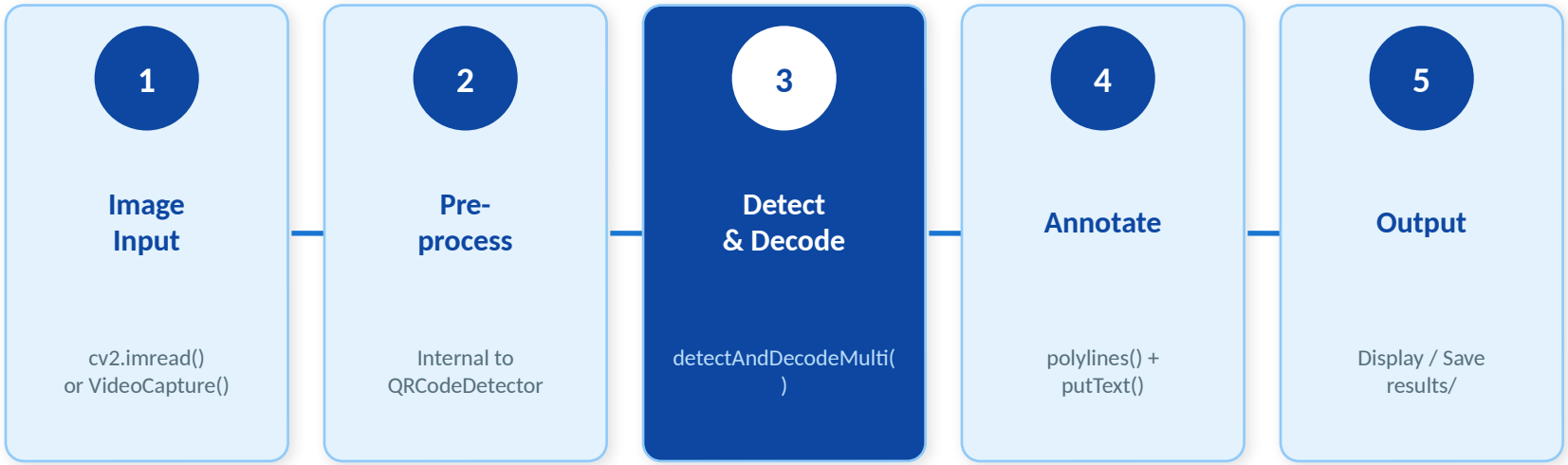
⚡

Real-time

Process ≥ 20 FPS on a standard CPU

Real-time

Detection Pipeline




```
Core: cv2.QRCodeDetector().detectAndDecodeMulti(frame)
```

Dataset Description

Category	# Images	Resolution	Conditions
Ideal QR codes	20	640×480	White background, upright
Rotated / skewed	15	640×480	Up to 45° tilt
Real-world photos	15	1280×720	Mixed lighting, distance
Total	**50**	**—**	**Custom dataset**

Images sourced from: self-generated QR codes (qr-code-generator.com) and real-world product photographs taken under varied conditions.

Experimental Setup & Code

💻 Language

Python 3.10

🎥 Camera

720p webcam

📦 Library

OpenCV 4.8

💾 RAM

8 GB

🖥️ CPU

Intel Core i5 (8th Gen)

```
detector = cv2.QRCodeDetector()
ok, decoded, points, _ = detector.detectAndDecodeMulti(frame)
```

Results & Evaluation Metrics

94%	27	~36ms	0%
Overall Accuracy	**FPS (real-time)**	**Avg Decode Time**	**False Positive Rate**
47 / 50 detected	Frames per second	Per image	Zero false alarms

Failure Analysis — 3 undetected images:

- Extreme motion blur (camera shake during capture)
- Very small codes < 40×40 pixels
- Partial occlusion of QR finder patterns

Conclusion & Future Work

✅ Conclusion

Successfully implemented a QR Code & Barcode Detection System with 94% accuracy and real-time 27 FPS performance — using only Python and OpenCV, no GPU required.

🤖 WeChatQRCode deep learning model for degraded images

📱 ZBar integration for 1-D barcodes (EAN-13, Code 128)

☁️ REST API deployment (FastAPI) for cloud scanning

🗄️ Database logging of scanned codes with timestamps

Thank You!

Questions & Discussion

👤 Ismoil Rahimov | ID: 230514

🎓 Module: Image Processing — CAU Spring 2025–2026

⚙️ Stack: Python 3.10 · OpenCV 4.8 · NumPy

📊 Result: 94% accuracy · 27 FPS real-time